

CANNBench: four operators, three arms

2026-09-03 · official-tasks v1.1.1 · private 950PR runs

Runtime pin: compiler-team/pyasc v2@030e9b2c0ce44cbc5f9523e03e131f4a23c23a2d
Operators: gelu, rms_norm, softmax, transpose
Verified skill loads: 24/24

Experimental design

Handwritten target-derived kernels are copied from `python/test/asctile/target` with only generated or deterministic host transport adaptation. The no-skill arm exposes no skill path and rejects skill calls. The with-skills arm requires native OpenCode skill calls in every accepted phase. CANNBench sees only the resulting wheel, so repository evidence establishes generation provenance.

Controlled exceptions: skill-guided Softmax reviewed the no-skill 20/20 baseline and retained the same hash. Transpose generation repeatedly exhausted model context before writing; both comparison arms therefore use the same target-derived candidate with deterministic rank-5/int64 remediation and differ only in their recorded review path. A Transpose performance delta cannot be attributed to skills.

Generated-arm provenance

Arm	Operator	Candidate hash	Phase models	Session IDs	Verified/required	Skill calls
Generated without skills	gelu	d74d059af15df90e	dashscope/qwen3.7-max, dashscope/qwen3.7-max, dashscope/qwen3.7-max	ses_f9a1c4471ffe6qXdKZZUYiszJf, ses_f9a0c38c3ffealzKwgHGwsLImm, ses_f9a095e94ffeVo2Rhbj9BIK4VZ	0/0	0
Generated without skills	rms_norm	d99df6e42d5a7959	dashscope/glm-5.2, dashscope/glm-5.2, dashscope/qwen3.7-max	ses_f9a237f9effeQ6xafR3AexDToz, ses_f9a1e33dffehSJZJCw0oNuDu6, ses_f9a1d603affeGBmzoSXP98MIXW	0/0	0
Generated without skills	softmax	ce232b6d4b938667	dashscope/qwen3.7-max, dashscope/qwen3.7-max, dashscope/qwen3.7-max	ses_f99dd6d0affea4eIRvuWCa9Q75, ses_f99d158bcfe8Dq6iUihvUJOso, ses_f99cbf58dffeAgaNPH0Up13h6K	0/0	0
Generated without skills	transpose	887e9d7cca7134bb	dashscope/qwen3.7-max, deterministic:human, dashscope/qwen3.7-max	ses_f99fbaf37ffeyhT3mhaOoAkwvi, —, ses_f99916d81ffeccorcZm6lu813n	0/0	0
Generated with skills	gelu	9e962802b651f836	dashscope/glm-5.2, dashscope/glm-5.2, dashscope/qwen3.7-max	ses_f9a057a23ffe1W8pco4Rf1AU7Y, ses_f99fab716ffeq0B6GjWP1OLat4, ses_f99f3c9e3ffeM1e1TVka79xfnV	7/7	7
Generated with skills	rms_norm	8f976fc4c9c5d00e	dashscope/qwen3.7-max, dashscope/qwen3.7-	ses_f99fd30d7ffe5fAC5Qst8KhLJ4, ses_f99e91256ffe8Psm8R1O3Z6opl, ses_f99e5abfcfe9B1lgU7qHqYqLb	7/7	7

Arm	Operator	Candidate hash	Phase models	Session IDs	Verified/required	Skill calls
			max, dashscope/qwen3.7-max			
Generated with skills	softmax	ce232b6d4b938667	dashscope/qwen3.7-max, deterministic:no_skills-candidate, dashscope/qwen3.7-max	ses_f99d5421dffe0KcbdYncuWXsBc, —, ses_f999f55c4ffeeMGESc0N4F7vce	5/5	5
Generated with skills	transpose	887e9d7cca7134bb	dashscope/qwen3.7-max, deterministic:no_skills-remediated-candidate, dashscope/qwen3.7-max	ses_f99df7100ffeXn9a0Ncz6iG1nH, —, ses_f99910755ffeWC7Gx2oAd1K4kO	5/5	5

Local compile gate

Arm	Operator	Dispatch	Compile	Status
Handwritten target-derived	gelu	14/20	14/20	failed
Handwritten target-derived	rms_norm	20/20	20/20	passed
Handwritten target-derived	softmax	14/20	14/20	failed
Handwritten target-derived	transpose	17/20	17/20	failed
Generated without skills	gelu	20/20	20/20	passed
Generated without skills	rms_norm	20/20	20/20	passed
Generated without skills	softmax	20/20	20/20	passed
Generated without skills	transpose	20/20	20/20	passed
Generated with skills	gelu	20/20	20/20	passed
Generated with skills	rms_norm	20/20	20/20	passed
Generated with skills	softmax	20/20	20/20	passed
Generated with skills	transpose	20/20	20/20	passed

The QEMU gate checks dispatch, pyasc lowering, AscendC translation and UB limits, but not numerical execution or latency. Any worker prose that called this `verified-cannbench` is rejected; only a completed remote job receives that label.

CANNBench results

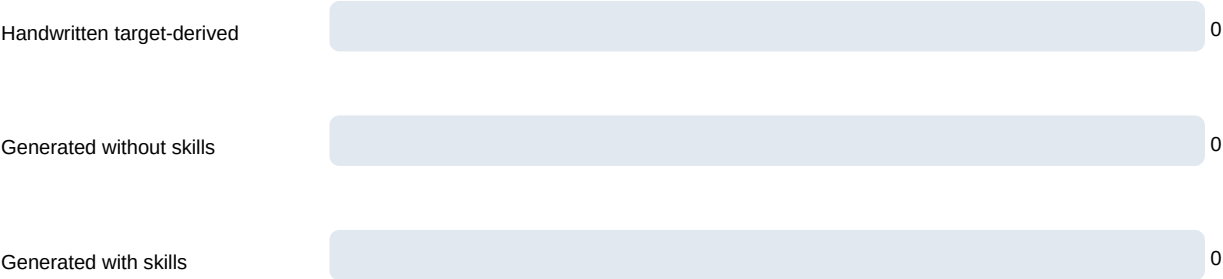
Run pages

- Handwritten target-derived: [job_00385e9c7a19](#)

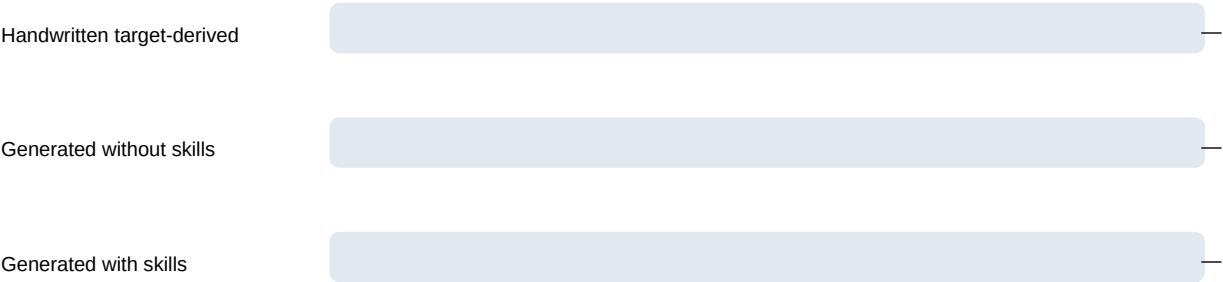
Arm	Job	Cases	Score	GMean	Anti-cheat	Status
Handwritten target-derived	job_00385e9c7a19	—	—	—	—	queued

Arm	Job	Cases	Score	GMean	Anti-cheat	Status
Generated without skills	pending	—	—	—	—	pending
Generated with skills	pending	—	—	—	—	pending

Passed cases (maximum 80)



Geometric-mean speedup



Interpretation

Correctness is a hard gate: partial-pass speedups are not comparable with complete solutions. The target-derived arm measures upstream target coverage of the wider CANNBench contract; it is not silently rewritten. Generated arms retain separate skill/no-skill provenance.

Evidence

- MANIFEST.json — hashes and runtime pin
- generated/*/*/evidence — worker events, sessions and skill traces
- handwritten/*/provenance.json — source definition hashes
- */local_validation — QEMU reports
- remote_runs — jobs and logs
- submissions/*.zip — immutable archives